LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 11 Test (GSEB pattern): Electricity

Chapter / Topic: Electricity

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. A wire of resistance R is cut into five equal parts, which are then connected in parallel. If the equivalent resistance is R′, the ratio R/R′ is: **[1]**

(a) 1/25 (b) 1/5
(c) 5 (d) 25

Q2. The SI unit of potential difference is the: **[1]**

(a) volt (b) ampere
(c) ohm (d) coulomb

Q3. Which of the following does NOT represent electric power in a circuit? **[1]**

(a) I^2R (b) IR^2
(c) VI (d) V^2/R

Q4. The length of a wire is doubled and its area of cross-section is halved. Its resistivity: **[1]**

(a) doubles (b) halves
(c) becomes four times (d) stays the same

Q5. The SI unit of electric charge is the: **[1]**

(a) coulomb (b) ampere
(c) volt (d) ohm

Q6. Two 4 Ω resistors are connected first in series and then in parallel. The ratio of the series resistance to the parallel resistance is: **[1]**

(a) 1 (b) 2
(c) 4 (d) 8

Q7. How much current does an 1100 W appliance draw from a 220 V supply? **[1]**

(a) 0.5 A (b) 2 A
(c) 5 A (d) 10 A

Q8. How much electrical energy does a 100 W bulb use in 10 hours? **[1]**

(a) 0.1 kWh (b) 1 kWh
(c) 10 kWh (d) 100 kWh

Q9. An electric bulb is rated 220 V, 100 W. When it is operated on 110 V, the power consumed is: **[1]**

(a) 25 W (b) 50 W
(c) 75 W (d) 100 W

SECTION B

6 marks

Q10. How much work is done in moving a charge of 2 C across two points that have a potential difference of 12 V? **[2]**

Q11. A current of 0.5 A flows through a bulb for 10 minutes. How much charge flows through it? **[2]**

Q12. Resistors of 4 Ω and 6 Ω are connected in series to a 12 V battery. Find the current in the circuit and the potential difference across the 4 Ω resistor. **[2]**

SECTION C

6 marks

Q13. A copper wire has a diameter of 0.5 mm and a resistivity of 1.6×10^{-8} Ω m. **[3]**

(i) What length of this wire will have a resistance of 10 Ω? **[2]**

(ii) If the diameter is doubled (same length), how does the resistance change? **[1]**

Q14. Resistors of 2 Ω, 3 Ω and 5 Ω are connected in series to a 10 V battery. **[3]**

(i) Find the total resistance. **[1]**

(ii) Find the current in the circuit. **[1]**

(iii) Find the potential difference across the 5 Ω resistor. **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. In an experiment, a student measured the current (I) through a resistor for different potential differences (V). The readings were: I = 0.5, 1.0, 2.0, 3.0, 4.0 A and V = 1.6, 3.4, 6.7, 10.2, 13.2 V respectively. **[4]**

(i) What relationship between V and I do these readings show? **[1]**

(ii) Calculate the resistance using the reading I = 2.0 A. **[1]**

(iii) The resistor is replaced by a wire of the same material and thickness but double the length. What current will flow at 6.7 V? **[2]**

~ All the best! ~